Nested Subqueries

- SQL provides a mechanism for the nesting of subqueries. A **subquery** is a **select-from-where** expression that is nested within another query.
- The nesting can be done in the following SQL query

  select $A_1, A_2, \ldots, A_n$
  from $r_1, r_2, \ldots, r_m$
  where P

  as follows:

  - **From clause:** r_i can be replaced by any valid subquery
  - **Where clause:** P can be replaced with an expression of the form:

    B <operation> (subquery)

    B is an attribute and <operation> to be defined later.
  - **Select clause:**

    A_i can be replaced be a subquery that generates a single value.

Vikram Pudi
Professor

Deletion

- Delete all instructo

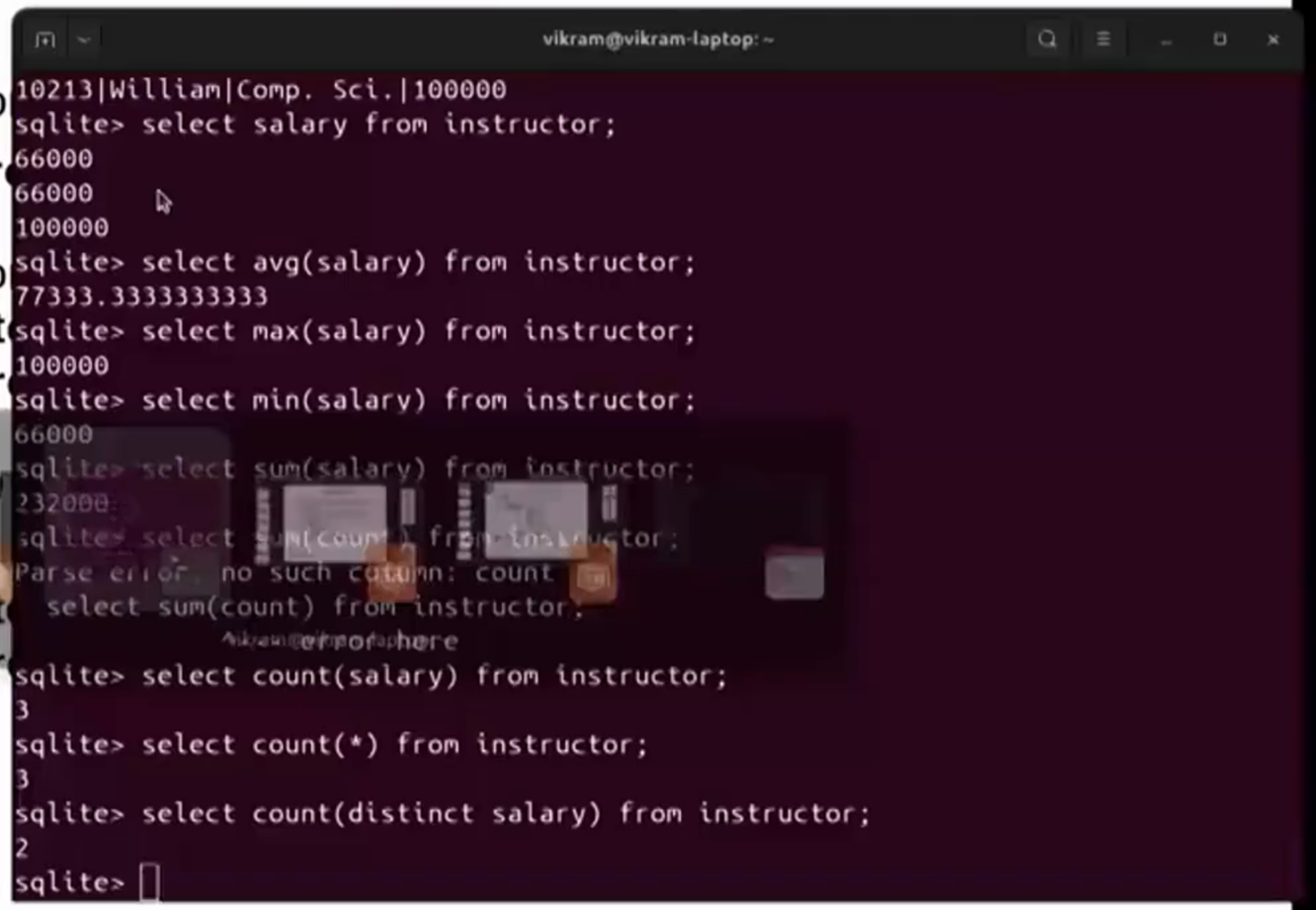
  delete fr

- Delete all instructo

  delet
  wher

- *Delete all tuples in*
  with a depart

  delet
  wher





Deletion

- Delete all instructors

  delete from *instructor*

- Delete all instructors from the Finance department

  delete from *instructor*
  where *dept_name*= 'Finance';

- *Delete all tuples in the instructor relation for those instructors associated with a department located in the Watson building.*

  delete from *instructor*
  where *dept name* **in** (**select** *dept name*
  from *department*
  where *building* = 'Watson');



Insertion (Cont.)

- Make each student in the Music department who has earned more than 144 credit hours an instructor in the Music department with a salary of $18,000
- SQL includes a string-matching operator for comparisons on character strings. The operator "like" uses patterns that are described using two special characters:
  - percent (%). The % character matches any substring.
  - underscore (_). The _ character matches any character.
- Find the names of all instructors whose name includes the substring "dar".
- Match the string "100%" using the like clause: like '100 %' escape '\' (would cause problem in that above w

insert into instructor
select ID, name, dept_name, 18000
from student
where dept_name = 'Music' and total_cred > 144;

The select from where statement is evaluated fully before any of its results are inserted into the relation.

Otherwise, insert statements like

insert into *table1* select * from *table1*

```
                error here ---^
sqlite> .schema instructor
CREATE TABLE instructor (
        ID              char(5),
        name            varchar(20) not null,
        dept_name       varchar(20),
        salary          numeric(8,2),
        primary key (ID),
        foreign key (dept_name) references department);
sqlite> delete from instructor where dept_name='Biology';
sqlite> select * from instructor;
10212|Jones|Physics|66000
10213|William|Comp. Sci.|100000
sqlite> .tables
course      instructor  student
sqlite> .schema course
CREATE TABLE course (
        course_id       varchar(8),
        title           varchar(50),
        dept_name       varchar(20),
        credits             numeric(2,0),
        primary key (course_id),
        foreign key (dept_name) references department);
sqlite>
```

Insertion (Cont.)

- Make each student in the Music department who has earned more than 144 credit hours an instructor in the Music department with a salary of $18,000.

  insert into *instructor*
  select *ID, name, dept_name, 18000*
  from
  where

- The **select from where** statement is evaluated fully before any of its results are inserted into the relation.

Otherwise queries like

insert into *table1* **select * from** *table1*

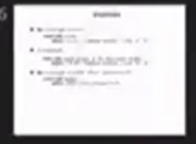
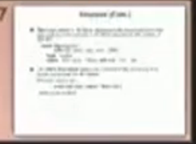
would cause problem

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- The **select from where** statement is evaluated fully before any of its results are inserted into the relation.

Otherwise queries like

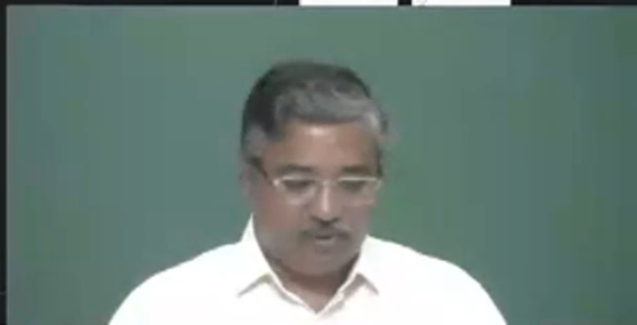
insert into *table*1 **select * from** *table*1

would cause problem

3.57







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Definition of "some" Clause

- F <comp> **some** $r \Leftrightarrow \exists\ t \in r$ such that (F <comp> t)
  Where <comp> can be: $<$, $\leq$, $>$, $=$, $\neq$

(5 < **some** | 0 / 5 / 6 |) = true (read: 5 < some tuple in the relation)

(5 < **some** | 0 / 5 |) = false

(5 = **some** | 0 / 5 |) = true

(5 ≠ **some** | 0 / 5 |) = true (since 0 ≠ 5)

(= **some**) ≡ **in**

However, (≠ **some**) ≢ **not in**

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Set Comparison – "some" Clause

- Find names of instructors with salary greater than that of some (at least one) instructor in the Biology department.

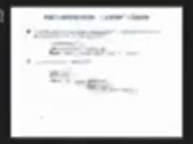
```
select distinct T.name
from instructor as T, instructor as S
where T.salary > S.salary and S.dept name = 'Biology';
```

- Same query using > **some** clause

```
select name
from instructor
where salary > some (select salary
                     from instructor
                     where dept name = 'Biology');
```


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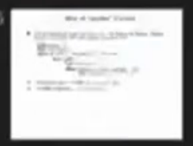










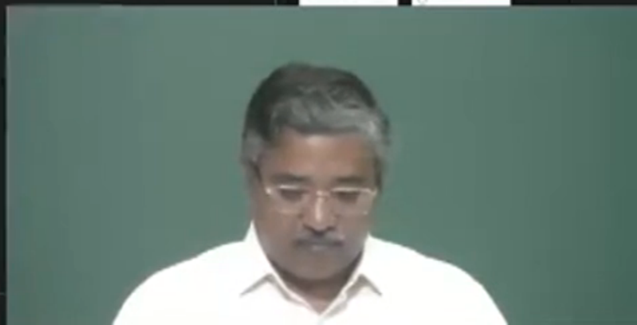




Set Comparison

3.39





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Set Comparison – "some" Clause

- Find names of instructors with salary greater than that of some (at least one) instructor in the Biology department.

```
select distinct T.name
from instructor as T, instructor as S
where T.salary > S.salary and S.dept name = 'Biology';
```

- Same query using > **some** clause

```
select name
from instructor
where salary > some (select salary
                     from instructor
                     where dept name = 'Biology');
```

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Set Comparison – "all" Clause

- Find the names of all instructors whose salary is greater than the salary of all instructors in the Biology department.

```
select name
from instructor
where salary > all (select salary
                    from instructor
                    where dept name = 'Biology');
```

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Complex Queries using With Clause

- Find all departments where the total salary is greater than the average of the total salary at all departments

```
with dept_total (dept_name, value) as
        (select dept_name, sum(salary)
         from instructor
         group by dept_name),
dept_total_avg(value) as
        (select avg(value)
        from dept_total)
select dept_name
from dept_total, dept_total_avg
where dept_total.value > dept_total_avg.value;
```

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Scalar Subquery

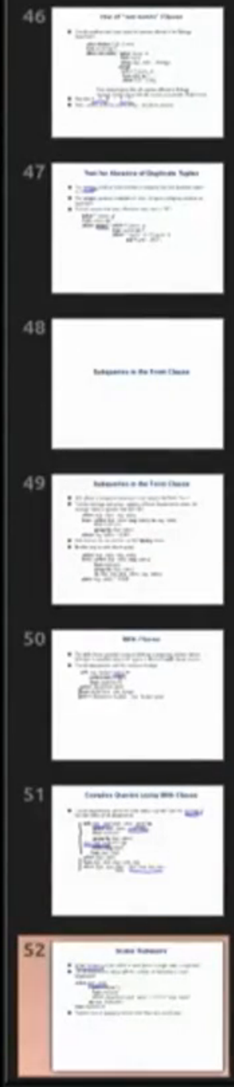
- Scalar subquery is one which is used where a single value is expected
- List all departments along with the number of instructors in each department

```
select dept_name,
        ( select count(*)
          from instructor
          where department.dept_name = instructor.dept_nam
        as num_instructors
from department;
```

- Runtime error if subquery returns more than one result tuple

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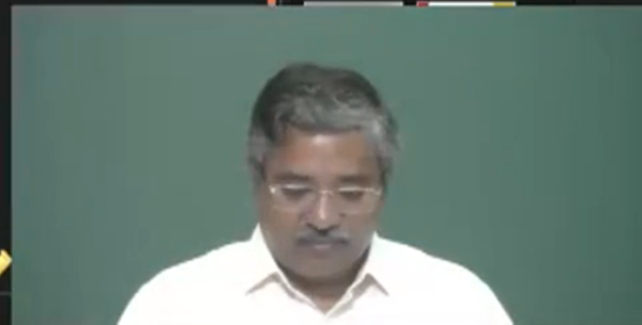


```
            from instructor
            where department.dept_name = instructor.dept_name)
        as num_instructors
from department;
```

- Runtime error if subquery returns more than one result tuple







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Modification of the Database

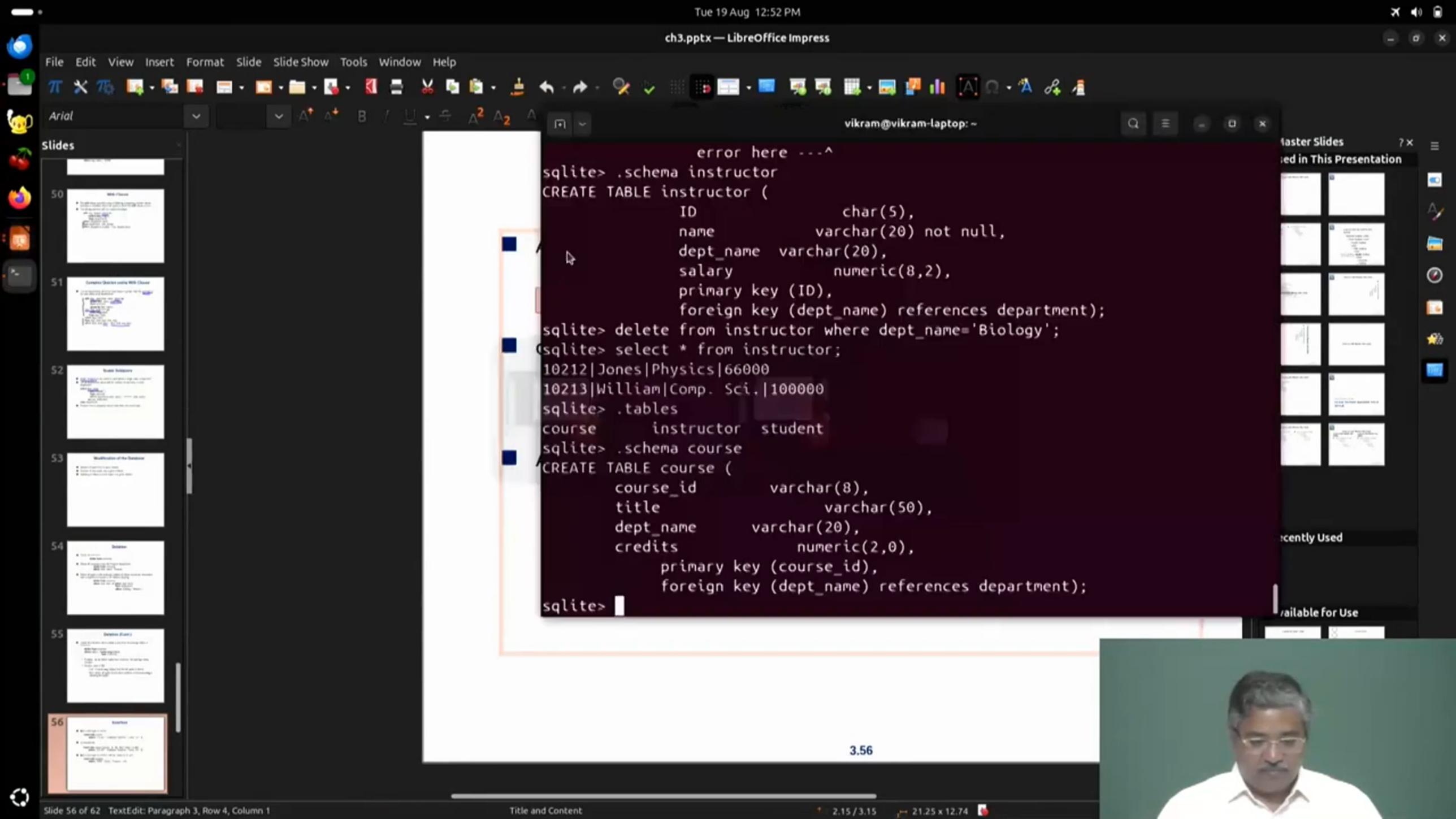
- Deletion of tuples from a given relation.
- Insertion of new tuples into a given relation
- Updating of values in some tuples in a given relation

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vikram@vikram-laptop: ~
error here ---^
sqlite> .schema instructor
CREATE TABLE instructor (
ID char(5),
name varchar(20) not null,
dept_name varchar(20),
salary numeric(8,2),
primary key (ID),
foreign key (dept_name) references department);
sqlite> delete from instructor where dept_name='Biology';
sqlite> select * from instructor;
10212|Jones|Physics|66000
10213|William|Comp. Sci.|100000
sqlite> .tables
course instructor student
sqlite> .schema course
CREATE TABLE course (
course_id varchar(8),
title varchar(50),
dept_name varchar(20),
credits numeric(2,0),
primary key (course_id),
foreign key (dept_name) references department);
sqlite>
3.56
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Insertion

- Add a new tuple to *course*

```
insert into course
    values ('CS-437', 'Database Systems', 'Comp. Sci.', 4);
```

- or equivalently

```
insert into course (course_id, title, dept_name, credits)
    values ('CS-437', 'Database Systems', 'Comp. Sci.', 4);
```

- Add a new tuple to *student* with *tot_creds* set to null

```
insert into student
    values ('3003', 'Green', 'Finance', null);
```


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